6.2202 class notes

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1 Circuit review

Laws. KCL: at every node, $\sum$(currents in) = $\sum$(currents out). KVL: around every closed loop, $\sum V = 0$.

Phasors. Sinusoid $v(t) = V_{max}\cos(\omega t + \phi)$. Positive ϕ shifts the curve left, so the waveform occurs earlier. RMS value: $\tilde{V} = V_{rms} = V_{max}/\sqrt{2}$. Power engineering reports V_{rms}, not V_{max}; state which you use. The phasor of $v(t) = \mathrm{Re}\{\tilde{V}\cos(\omega t+\phi) + j\tilde{V}\sin(\omega t+\phi)\} = \mathrm{Re}\{\tilde{V}e^{\omega t+\phi}\}$. Write as phasor: $V = \tilde{V}\angle\phi$, a complex number with RMS magnitude and angle ϕ. $j = \sqrt{-1} = 1\angle 90°$, and $1/j = -j$. Rectangular $\iff$ polar: $a + jb = \sqrt{a^2+b^2}\angle\,\mathrm{atan2}(b,a)$, and $r\angle\theta = r\cos\theta + jr\sin\theta = re^{j\theta}$. Add and subtract in rectangular. Multiply and divide in polar: $(r_1\angle\theta_1)(r_2\angle\theta_2) = r_1r_2\angle(\theta_1+\theta_2)$, $(r_1\angle\theta_1)/(r_2\angle\theta_2) = (r_1/r_2)\angle(\theta_1-\theta_2)$. Conjugate: $(r\angle\theta)^* = r\angle(-\theta)$.

Elements. Sign convention: I is defined entering the + terminal of V. Impedance $Z = R + jX$ [Ω], with resistance R and reactance X. Polar: $Z = |Z|\angle\theta$, $|Z| = \sqrt{R^2+X^2}$, $\tan\theta = X/R$. Admittance $Y = 1/Z = G + jB$ [S], with conductance G and susceptance B; in polar $Y = (1/|Z|)\angle(-\theta)$. Ohm's law: $V = IZ$, or $I = VY$. With $V = \tilde{V}\angle\phi_V$ and $I = \tilde{I}\angle\phi_I$, $\theta = \phi_V - \phi_I$. Inductor L: $v = L\,di/dt$, so $X_L = \omega L$. Capacitor C: $i = C\,dv/dt$, so $X_C = -1/(\omega C)$. Inductive ("lag"): $\theta > 0$, $\phi_I < \phi_V$, current occurs after voltage. Capacitive ("lead"): $\theta < 0$, $\phi_I > \phi_V$, current occurs before voltage.

Solving. Series: $Z_{eq} = Z_1 + Z_2 + \ldots$; same current through each. Parallel: $Y_{eq} = Y_1 + Y_2 + \ldots$, or $Z_{eq} = Z_1Z_2/(Z_1+Z_2)$ for two; same voltage across each. Voltage divider (series): $V_2 = V \cdot Z_2/(Z_1+Z_2)$. Current divider (parallel): $I_2 = I \cdot Z_1/(Z_1+Z_2)$. Thevenin: a source V_{Th} in series with Z_{Th}. V_{Th} is the open-circuit voltage at the terminals. Z_{Th} is the impedance seen at the terminals with independent sources zeroed (voltage source $\to$ short, current source $\to$ open), or $Z_{Th} = V_{oc}/I_{sc}$. Norton: a current source $I_N = V_{Th}/Z_{Th}$ in parallel with $Z_N = Z_{Th}$.

Transients. First-order RC or RL circuit with a DC source: $x(t \geq 0) = x(\infty) + [x(0^+) - x(\infty)]\,e^{-t/\tau}$ for any voltage or current x. Time constant $\tau = R_{Th}C$ (RC) or L/R_{Th} (RL), with R_{Th} the Thevenin resistance seen by v_C or i_L. v_C and i_L cannot jump: $x(0^+) = x(0^-)$. $x(\infty)$: capacitor open, inductor short. Any other variable at 0^+: redraw with C as a voltage source $v_C(0^+)$ (or L as a current source $i_L(0^+)$) and solve. Origin: particular solution (steady state) plus homogeneous $Ae^{-t/\tau}$, with A set by $x(0^+)$. Power absorbed by any element: $p(t) = v(t)\,i(t)$. Stored energy is $w(t) = \int_0^t p(t)\,dt = \int v(t)\,i(t)\,dt$. Capacitor: $i(t)\,dt = C\,dv$, so $w(t) = \int v(t)\,C\,dv = \frac{1}{2}Cv(t)^2$. Inductor: $v(t)\,dt = L\,di$, so $w(t) = \int L\,i(t)\,di = \frac{1}{2}Li(t)^2$.

2 Power

Definitions. Complex power $S = VI^*$ [VA]; S is power absorbed by the element. $S = P + jQ$, with real power P [W] and reactive power Q [VAr]. Polar: $S = \tilde{V}\tilde{I}\angle\theta$, where $\theta = \phi_V - \phi_I$. So $P = \tilde{V}\tilde{I}\cos\theta$, $Q = \tilde{V}\tilde{I}\sin\theta$, $|S| = \tilde{V}\tilde{I}$, and $Q = P\tan\theta$. Power factor pf $= \cos\theta = P/|S|$. $\cos\theta$ loses the sign of θ, so a pf is stated lagging or leading. Lagging: $\theta > 0$, inductive, $Q > 0$, absorbs Q. Leading: $\theta < 0$, capacitive, $Q < 0$, supplies Q. Single element: $S = \tilde{I}^2 Z = \tilde{V}^2 Y^*$.

Derivation of P. $p(t) = v(t)\,i(t) = V_{max}I_{max}\cos(\omega t + \phi_V)\cos(\omega t + \phi_I)$. By $\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha-\beta) + \cos(\alpha+\beta)]$, $p(t) = \frac{1}{2}V_{max}I_{max}[\cos(\phi_V - \phi_I) + \cos(2\omega t + \phi_V + \phi_I)]$. The first term is constant; the second averages to zero. So $P = \frac{1}{2}V_{max}I_{max}\cos\theta = \tilde{V}\tilde{I}\cos\theta$.

Load to element. A problem states a load as $|S|$ (or P) and a pf, lagging or leading. $\theta = \pm\arccos(\text{pf})$, positive if lagging. As a current: $I = (S/V)^*$, so $\tilde{I} = |S|/\tilde{V}$ and $\phi_I = \phi_V - \theta$. As an impedance: $Z = \tilde{V}^2/S^*$.

Conservation. $\sum S = 0$ at a node and P and Q balance separately. Power-factor correction: given P, pf θ_1, bus $V = \tilde{V}\angle 0$, target θ_2; find C in parallel with the load. $S_L = P + jP\tan\theta_1$; $S_C = |V|^2 Y^* = -j\omega C\tilde{V}^2$; $S_{src} = S_L + S_C = P + j(P\tan\theta_1 - \omega C\tilde{V}^2)$. Target: $Q_{src} = P\tan\theta_2$, since $Q = P\tan\theta$ and P is unchanged.

3 Three-phase and per unit

Definitions. Balanced: phase b is phase a $\angle -120°$, phase c is phase a $\angle +120°$, for every voltage and current; solve phase a only. Balanced line currents sum to zero, so the neutral carries no current and may be omitted. V_{ln}: phase to neutral, e.g. V_{an}. V_{ll}: phase to phase, $V_{ab} = V_{an} - V_{bn} = \sqrt{3}\,V_{an}\angle +30°$. Voltage ratings are V_{ll} unless stated. Relationship holds for both connection types. Line current: in a conductor between source and load (I_a). Phase current: in one load element; equals the line current in Y (I_{ab} in Δ).

Connections. Y: element between phase and neutral; element voltage V_{ln}, element current $I_a = V_{an}/Z_Y$. Δ: element between two phases; element voltage V_{ll}, element current $I_{ab} = V_{ab}/Z_\Delta$. Δ line current: $I_a = I_{ab} - I_{ca} = \sqrt{3}\,I_{ab}\angle -30°$. Δ to Y: $Z_Y = Z_\Delta/3$ gives the same line currents.

Solving. Convert every Δ to Y and every V_{ll} to $V_{ln} = V_{ll}/\sqrt{3}\angle -30°$. Draw phase a: source V_{an}, series source and line impedances, Y-equivalent load, neutral return (zero impedance, present or not); shunt elements line to neutral. A load given as $|S_{3\phi}|$ (or P) and a pf at load voltage V_{ll}: $S_\phi = |S_{3\phi}|\angle\theta/3 = P/(3\text{pf})\angle\theta$ with $\theta = \pm\arccos(\text{pf})$ positive if lagging; $Z_Y = \tilde{V}_{ln}^2/S_\phi^*$.

Power. $S_{3\phi} = 3V_{an}I_a^* = \sqrt{3}\,\tilde{V}_{ll}\tilde{I}_{line}\angle\theta$. Y: $|S|_{3\phi} = 3\tilde{V}_{ln}^2/|Z_Y|$. Δ: $|S|_{3\phi} = 3\tilde{V}_{ll}^2/|Z_\Delta| = 9\tilde{V}_{ln}^2/|Z_\Delta|$.

Per unit. One base power S_b for the network, three-phase, typically 100 MVA; one base voltage V_b per zone, line-to-line, nominal. pu = actual/base; scales magnitues only and works in polar and rectangular forms. Single-phase $I_b = S_b/V_b$, $Z_b = V_b^2/S_b$. Three-phase $V_{b,ln} = V_{b,ll}/\sqrt{3}$, $I_b = S_b/(\sqrt{3}V_{b,ll})$, $Z_b = V_{b,ll}^2/S_b$; Base change: $Z_{pu,new} = Z_{pu,old}\,(V_{b,old}/V_{b,new})^2\,(S_{b,new}/S_{b,old})$. In pu, $|V_{ln}| = |V_{ll}|$ and $S_\phi = S_{3\phi}$: the bases carry the $\sqrt{3}$ and the 3. Across a transformer, set the other side's V_b by its line-to-line ratio; then Z_{pu} is the same on both sides and the ideal transformer drops out.

4 Power flow and admittance matrices

Admittance matrix. Recipe: number the nodes, ground as reference; generators as Norton current injections I_k; impedance loads and line charging as shunt admittances to ground; series impedances as $y_{km} = 1/Zkm$; KCL at each node in node voltages, $I_k = \sum_m Y_{km}V_m$; so $I = YV$ and solve for V in complex arithmetic; branch current $y_{km}(V_k - V_m)$. Y_{kk}: sum of all admittances touching node k, series and shunt. Y_{km}: minus the series admittance between k and m; zero if none. Y is symmetric and is singular without a shunt. $Y = A^\top \operatorname{diag}(y_{\text{series}})A + \operatorname{diag}(y_{\text{shunt}})$, A the branch-to-node incidence matrix.

Power flow. $S_k = V_k I_k^* = V_k\sum_m Y_{km}^* V_m^*$, generation minus load. With $V_k = \tilde{V}_k\angle\phi_k$, $Y_{km} = |Y_{km}|\angle\delta_{km}$: $P_k = \sum_m \tilde{V}_k\tilde{V}_m|Y_{km}|\cos(\phi_k - \phi_m - \delta_{km})$. Each bus: P, Q, $\tilde{V}$, ϕ; $2N$ equations, $4N$ unknowns. PQ (load): P, Q given. PV (generator): P, $\tilde{V}$ given. Slack (one): $\tilde{V}$, $\phi = 0$ given; supplies remaining P including P_{loss}. Solve: Newton on $x = [\phi$ (PQ, PV); $\tilde{V}$ (PQ)] with mismatch $f(x) = [P_k(x) - P_k^{given};\ Q_k(x) - Q_k^{given}]$ at those buses: $x \leftarrow x - J^{-1}f(x)$ from $\tilde{V} = 1$, $\phi = 0$. $J = [\partial P/\partial\phi,\ \partial P/\partial\tilde{V}\ ;\ \partial Q/\partial\phi,\ \partial Q/\partial\tilde{V}]$, by differentiating the P_k, Q_k sums. Then slack P, Q and PV-bus Q from their equations. $P_{loss} = \sum_k P_k = \sum_{(k,m)\in\text{lines}}|I_{km}|^2 R_{km}$, with $I_{km} = y_{km}(V_k - V_m)$.

5 Power line models

Models. Per unit length: series r, x [Ω/km]; shunt b [S/km], line charging, capacitive. Length ℓ: $Z = (r + jx)\ell$, $Y = jb\ell$. Short line: Z only. π model: Z in series, $Y/2$ to neutral at each end. All values per phase; the line is one branch of the per-phase circuit. Ampacity: maximum safe $|I|$.

Setup. Sending end S (source), receiving end R (load): V_S, I_S, V_R, I_R, all line-to-neutral and line current. Rated voltage: $\tilde{V}_R$ at nominal, given as V_{ll} unless stated. Full load: I_R from the rated $|S|$ and pf at V_R; no load: $I_R = 0$. Solve from R back to S. Short line: $I_S = I_R$, $V_S = V_R + I_R Z$, $P_{loss} = 3|I_R|^2 R$. π model: $V_S = V_R + (I_R + V_R Y/2)Z$, $I_S = I_R + V_R Y/2 + V_S Y/2$. $P_{loss} = P_S - P_R$, with $P_S = 3\,\mathrm{Re}(V_S I_S^*)$, $P_R = 3\,\mathrm{Re}(V_R I_R^*)$. $\eta = P_R/P_S$.

6 Magnetism and magnetic circuits

Magnetism. Magnetic field intensity H [A/m]. Permeability $\mu = \mu_r \mu_0$ [H/m]; $\mu_0 = 4\pi \times 10^{-7}$ H/m. Magnetic flux density $B = \mu H$ [T = Wb/m^2]. Magnetic flux $\Phi = BA$ [Wb] through a cross-section of area A [m^2]. Flux linkage $\lambda = N\Phi$ [Wb·turn] for a coil of N turns. Faraday: $v = d\lambda/dt = N\, d\Phi/dt$ [V]. Ampère: $\oint H\, dl = I_{enc}$, the current passing through the loop formed by the path. For the mean core path of length l [m] with uniform H, and a coil of N turns carrying i [A] threading the loop: $Hl = Ni$. Ni is the magnetomotive force (MMF) F [A·turn].

Magnetic circuit. Electric $\iff$ Magnetic: EMF V [V] $\iff$ MMF $F = Ni$ [A·turn]; current I [A] $\iff$ flux Φ [Wb]; resistance R [Ω] $\iff$ reluctance $\mathcal{R} = l/(\mu A)$ [A/Wb], for a segment of length l, area A, permeability μ; KVL $\iff$ $\sum F - \sum \mathcal{R}\Phi = 0$ around a loop (Ampère); KCL $\iff$ $\sum \Phi = 0$ at a junction (flux continuity); Ohm's law $V = IR$ $\iff$ $F = \Phi\mathcal{R}$; series and parallel combine like resistance. Coil = MMF source of value Ni, direction by the right-hand rule. One $\mathcal{R}$ per segment of distinct l, μ, or A. Air gap: $\mathcal{R}_{gap} = l_{gap}/(\mu_0 A)$ dominates, since $\mu_0 \ll \mu_{core}$.

Inductance. Coil of N turns on a single core of length l, area A: $L = \lambda/i = (N\Phi)/i = (N(BA))/i = (N((\mu H)A))/i = (N((\mu(Ni/l))A))/i = N^2/\mathcal{R}$ [H]. Several segments: $L = N^2/\mathcal{R}_{total}$, with $\mathcal{R}_{total}$ seen by the coil.

Saturation. B cannot exceed B_{sat} anywhere in the core. Procedure: $F = Ni_{peak}$; Φ from the magnetic circuit; $B_k = \Phi_k/A_k$ in each segment k; $B_{sat} \geq \max_k B_k$, in the smallest A_k carrying the largest Φ_k.

Losses. Core, eddy: flux induces voltage in the core, voltage drives current, current heats; $\propto f^2 B^2$; fix with laminations and silicon steel. Core, hysteresis: area of the B–H loop per cycle; $\propto f\times$ area; fix by material. Winding, skin: AC current crowds to the surface; fix with Litz wire. Winding, proximity: one conductor's field induces eddy currents in its neighbors, worse with more layers; fix with fewer or interleaved layers, Litz wire.

7 Transformers

$VR = (|V_{R,NL}| - |V_{R,FL}|)/|V_{R,FL}|$ at fixed V_S; short line $V_{R,NL} = V_S$, π model $V_{R,NL} = V_S/(1 + ZY/2)$.

8 Machines